

Redis Interview Preparation Guide

This guide covers important Redis concepts frequently asked in backend and full stack developer interviews. It includes explanations, use cases, interview questions, and practical examples.

1. Redis Data Structures

Redis supports multiple in-memory data structures:

Strings – Store text, JSON, counters, tokens.

Example: SET user:1 "Vignesh"

Lists – Ordered collection used for queues.

Example: LPUSH tasks "job1"

Sets – Unique values, useful for tags and unique users.

Example: SADD onlineUsers 101

Hashes – Store objects with key-value fields.

Example: HSET user:1 name "Vignesh" age 24

Sorted Sets (ZSET) – Sorted by score, useful for leaderboards.

Example: ZADD leaderboard 1000 player1

Important Interview Questions:

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| • What are the main Redis data structures? |
| • Difference between List and Set? |
| • Why use Hashes instead of Strings? |
| • What is Sorted Set used for? |

2. Redis Streams

Redis Streams is a log-based data structure used for event streaming and messaging systems.

Features:

- Message persistence
- Consumer groups
- Message acknowledgement
- Real-time processing

Common Use Cases:

- Chat applications
- Notification systems
- Order processing
- Event-driven architecture

Example:

```
XADD mystream * message "hello"
```

Important Interview Questions:

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| • What are Redis Streams? |
| • Difference between Pub/Sub and Streams? |
| • What are Consumer Groups? |
| • Why use Streams in distributed systems? |

3. Persistence (RDB & AOF)

Redis is in-memory but supports persistence.

RDB (Redis Database Backup)

- Snapshot-based persistence
- Saves data periodically
- Faster recovery

AOF (Append Only File)

- Logs every write operation
- Better durability
- Slower compared to RDB

Best Practice:

Use both RDB + AOF together in production.

Important Interview Questions:

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| • Difference between RDB and AOF? |
| • Which persistence is faster? |
| • Which provides better durability? |
| • Why use both RDB and AOF? |

4. LRU/LFU Eviction Policies

Redis removes keys when memory is full using eviction policies.

LRU (Least Recently Used)

Removes keys that were not recently accessed.

LFU (Least Frequently Used)

Removes keys accessed less frequently.

Common Policies:

- allkeys-lru
- volatile-lru
- allkeys-lfu
- noeviction

Important Interview Questions:

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| • What is cache eviction? |
| • Difference between LRU and LFU? |
| • Which eviction policy is best for caching? |
| • What happens when Redis memory becomes full? |

5. Redis Transactions

Redis transactions execute multiple commands atomically.

Commands:

- MULTI – Start transaction
- EXEC – Execute commands
- DISCARD – Cancel transaction
- WATCH – Optimistic locking

Example:

```
MULTI
SET user:1 "Vignesh"
INCR counter
EXEC
```

Important Interview Questions:

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| • How do Redis transactions work? |
| • What is WATCH command? |
| • Does Redis support rollback? |
| • Difference between Transactions and Lua scripts? |

Additional Redis Topics to Prepare:

- Pub/Sub
- Distributed Locking
- Rate Limiting
- Redis Cluster
- Replication
- Sentinel
- BullMQ Queues
- TTL Expiration
- Session Storage

- Caching Strategies